

In [2]: `import pandas as pd`

```
df = pd.read_csv("Superstore.csv", encoding="latin1")
df.head()
```

Out[2]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Co
0	1	CA- 2016- 152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	L
1	2	CA- 2016- 152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	L
2	3	CA- 2016- 138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	L
3	4	US- 2015- 108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	L
4	5	US- 2015- 108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	L

5 rows × 21 columns



In [3]:

```
df.info()

df.describe()

df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                 9994 non-null   int64
1   Order ID               9994 non-null   object
2   Order Date             9994 non-null   object
3   Ship Date              9994 non-null   object
4   Ship Mode              9994 non-null   object
5   Customer ID            9994 non-null   object
6   Customer Name          9994 non-null   object
7   Segment                9994 non-null   object
8   Country                9994 non-null   object
9   City                   9994 non-null   object
10  State                  9994 non-null   object
11  Postal Code            9994 non-null   int64
12  Region                 9994 non-null   object
13  Product ID             9994 non-null   object
14  Category               9994 non-null   object
15  Sub-Category           9994 non-null   object
16  Product Name           9994 non-null   object
17  Sales                  9994 non-null   float64
18  Quantity               9994 non-null   int64
19  Discount               9994 non-null   float64
20  Profit                 9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
Out[3]: Row ID           0
        Order ID        0
        Order Date      0
        Ship Date       0
        Ship Mode       0
        Customer ID     0
        Customer Name    0
        Segment         0
        Country         0
        City            0
        State           0
        Postal Code     0
        Region          0
        Product ID      0
        Category        0
        Sub-Category    0
        Product Name    0
        Sales           0
        Quantity        0
        Discount        0
        Profit          0
        dtype: int64
```

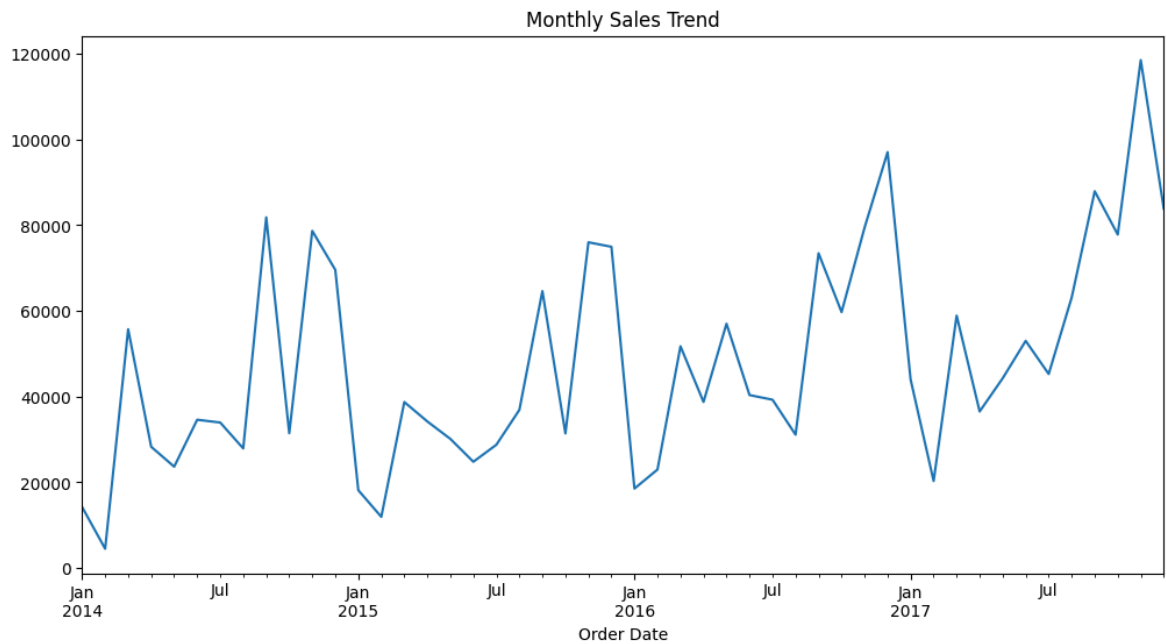
```
In [4]: df.drop_duplicates(inplace=True)

df['Order Date'] = pd.to_datetime(df['Order Date'])
```

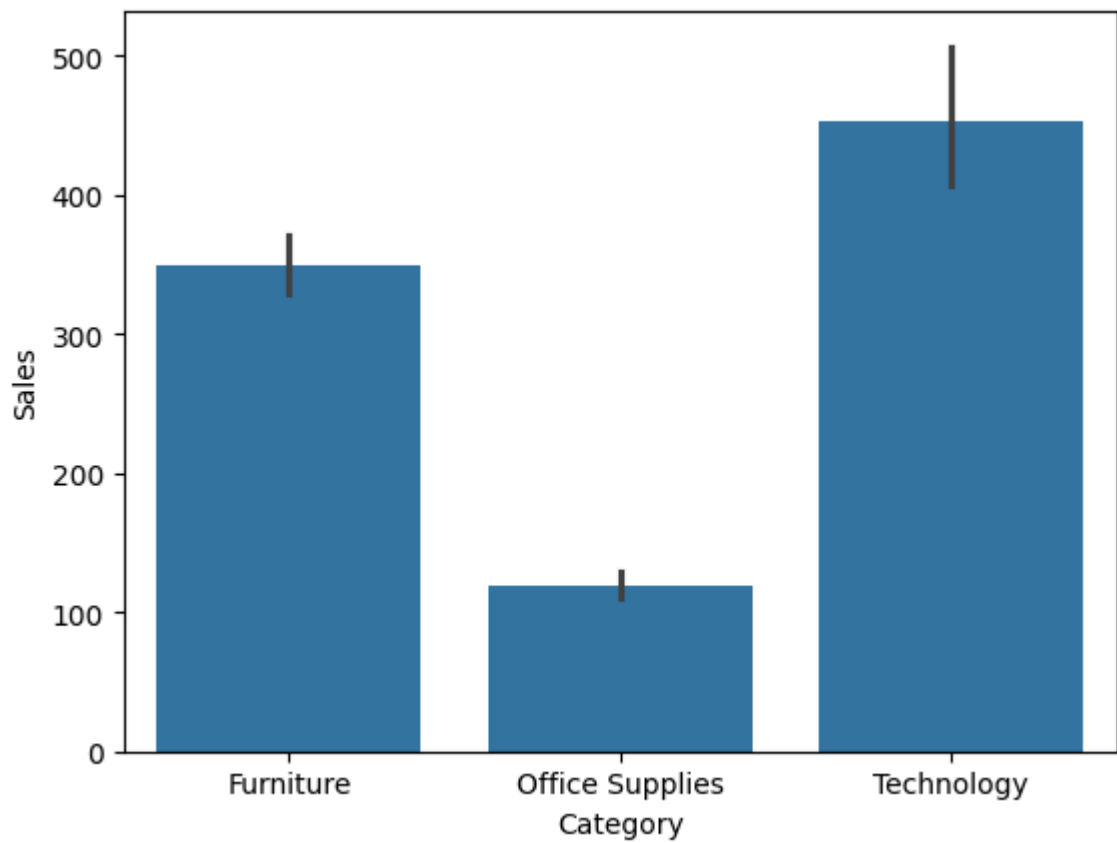
```
In [5]: monthly_sales = df.groupby(
        df['Order Date'].dt.to_period('M')
    )['Sales'].sum()
```

```
monthly_sales.plot(  
    figsize=(12,6),  
    title='Monthly Sales Trend'  
)
```

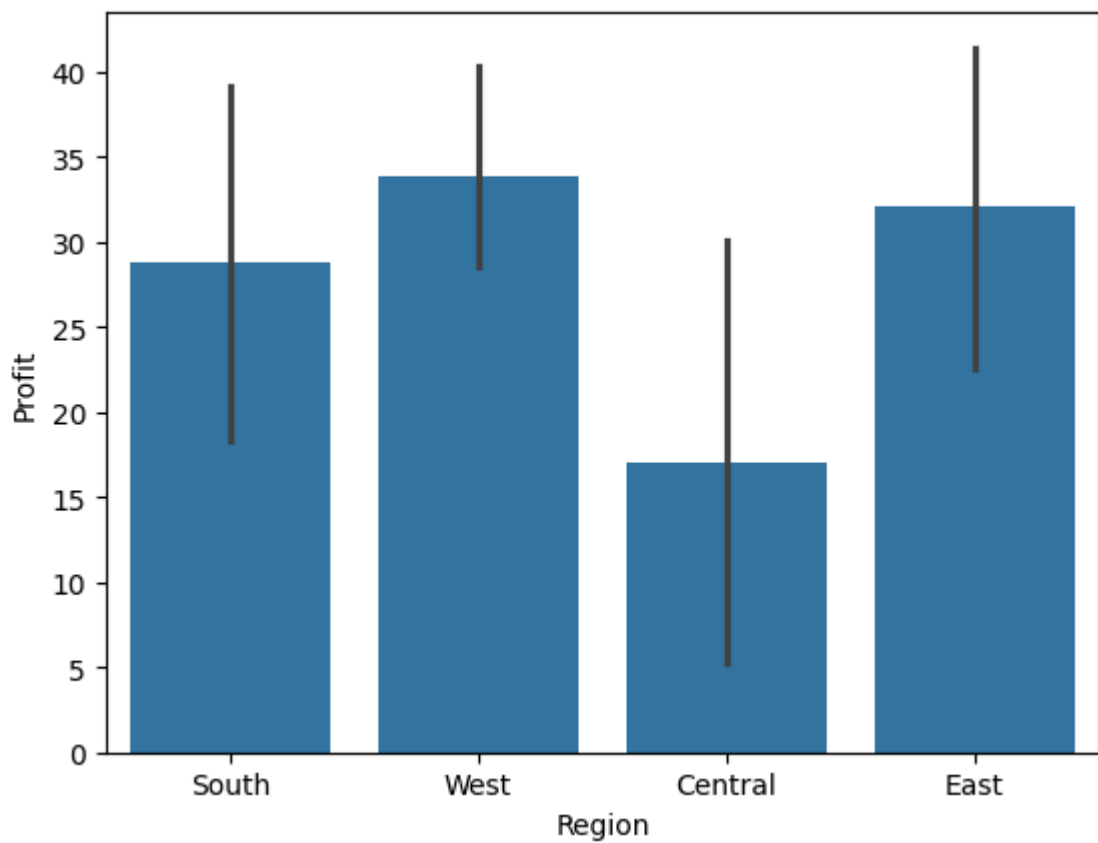
Out[5]: <Axes: title={'center': 'Monthly Sales Trend'}, xlabel='Order Date'>



```
In [6]: import seaborn as sns  
import matplotlib.pyplot as plt  
  
sns.barplot(  
    x='Category',  
    y='Sales',  
    data=df  
)  
  
plt.show()
```



```
In [7]: sns.barplot(  
    x='Region',  
    y='Profit',  
    data=df  
)  
  
plt.show()
```



```
In [8]: df.groupby('Category')['Sales'].sum().sort_values(ascending=False)
```

```
Out[8]: Category
Technology      836154.0330
Furniture       741999.7953
Office Supplies 719047.0320
Name: Sales, dtype: float64
```

```
In [9]: df.groupby('Region')['Profit'].sum().sort_values(ascending=False)
```

```
Out[9]: Region
West      108418.4489
East      91522.7800
South     46749.4303
Central   39706.3625
Name: Profit, dtype: float64
```

```
In [10]: top_products = df.groupby('Product Name')['Sales'].sum()

top_products.sort_values(ascending=False).head(10)
```

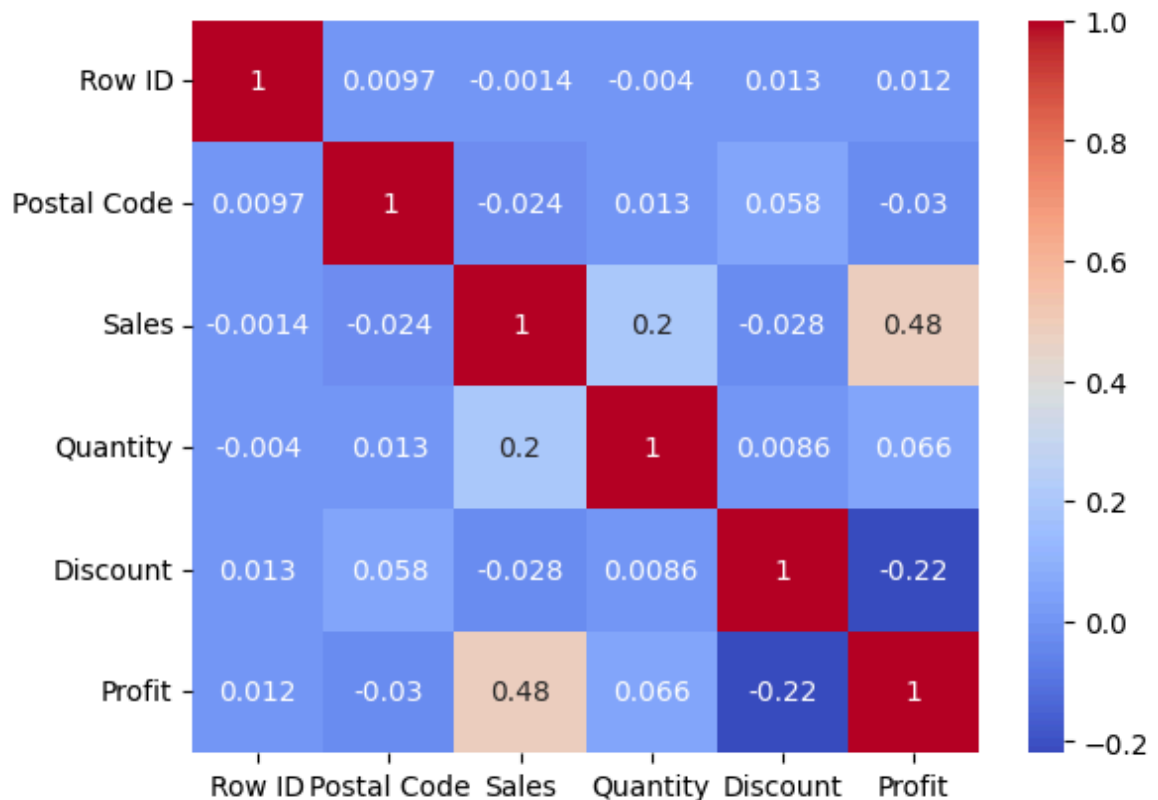
```
Out[10]: Product Name
Canon imageCLASS 2200 Advanced Copier
61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind
27453.384
Cisco TelePresence System EX90 Videoconferencing Unit
22638.480
HON 5400 Series Task Chairs for Big and Tall
21870.576
GBC DocuBind TL300 Electric Binding System
19823.479
GBC Ibimaster 500 Manual ProClick Binding System
19024.500
Hewlett Packard LaserJet 3310 Copier
18839.686
HP Designjet T520 Inkjet Large Format Printer - 24" Color
18374.895
GBC DocuBind P400 Electric Binding System
17965.068
High Speed Automatic Electric Letter Opener
17030.312
Name: Sales, dtype: float64
```

```
In [11]: numeric_cols = df.select_dtypes(include=['int64', 'float64'])

corr = numeric_cols.corr()

sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm'
)

plt.show()
```



```
In [12]: df['Month'] = df['Order Date'].dt.month
df['Year'] = df['Order Date'].dt.year
```

```
In [13]: X = df[['Month', 'Year', 'Quantity', 'Discount']]
y = df['Sales']
```

```
In [14]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
In [15]: from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

```
In [18]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

print("MAE:", mean_absolute_error(y_test, predictions))

rmse = np.sqrt(mean_squared_error(y_test, predictions))
print("RMSE:", rmse)

print("R2 Score:", r2_score(y_test, predictions))
```

MAE: 274.9499624219286
RMSE: 754.5124681893393
R2 Score: 0.0362453504268303

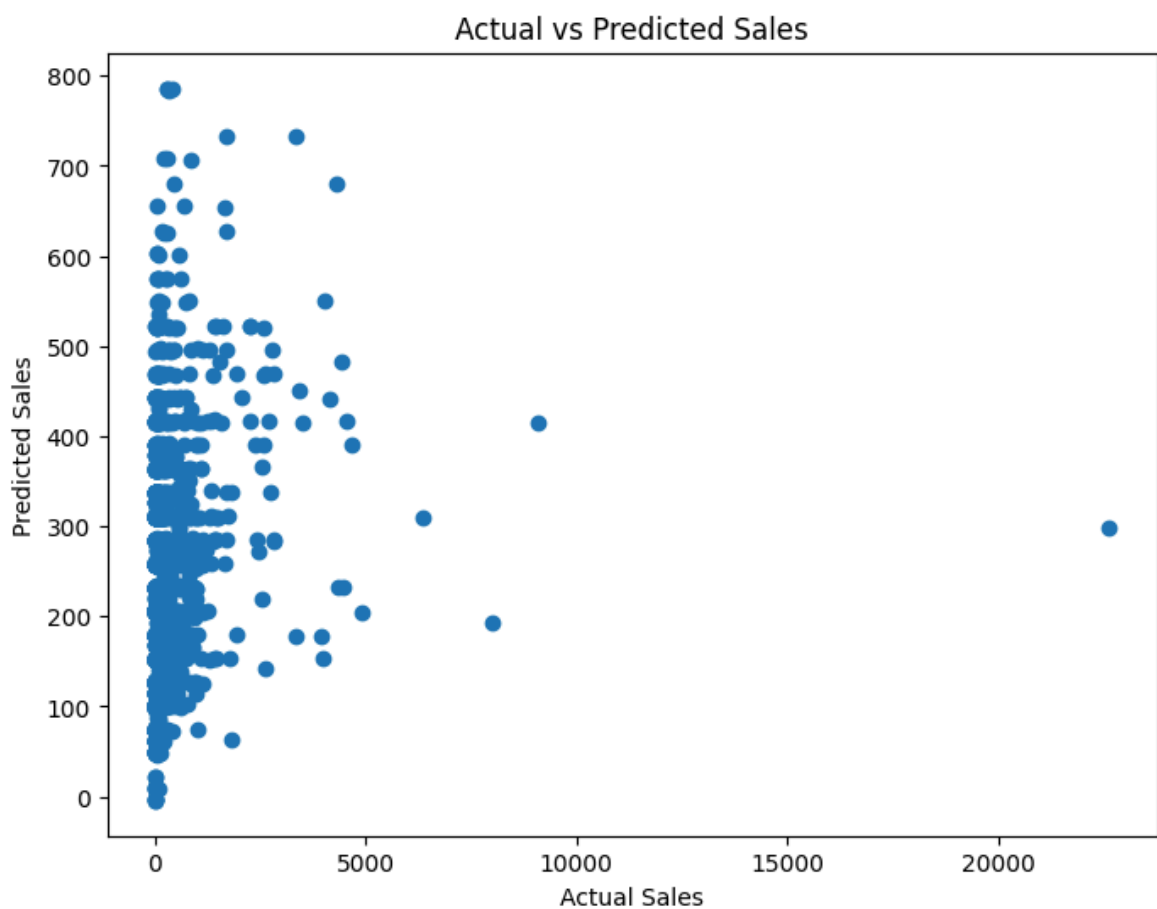
```
In [17]: plt.figure(figsize=(8,6))

plt.scatter(
    y_test,
    predictions
)

plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")

plt.title("Actual vs Predicted Sales")

plt.show()
```



In []: